

**Dr. B. S. Ratanpal**

**Assistant Professor**

Department of Applied Mathematics, Faculty of Technology & Engineering,  
The Maharaja Sayajirao University of Baroda, Vadodara – 390 001, Gujarat, India  
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**Date of Birth:**

30<sup>th</sup> August 1979

**Residential Address:**

A-9/35, Sant Kanwar Colony, Warashiya, Vadodara – 390 006, Gujarat, India

**Specialization:**

General Theory of Relativity, Orbital Dynamics.

**Scopus Author Id:**

8591657700

**Orchid Profile:**

<https://orcid.org/0000-0002-8593-3349>

**Subjects Taught:**

MATLAB, Python, Mechanics, Simulation and Modeling, Discrete Mathematics (Mathematical Logic and Theory of Algorithms), Image Processing, Laplace Transform, Fourier Series, Differential Equations.

**Educational Qualifications:**

1. Ph.D in Applied Mathematics from The Maharaja Sayajirao University of Baroda

**Title:** Application of Non-Linear Differential Equations in the Study of Superdense Stars on Geometrically Significant Spacetimes.

**Guide:** Prof. S. Ramamohan

2. M. Sc. (Applied Mathematics) from The Maharaja Sayajirao University of Baroda with First Class (64.75%).

3. B. Sc. (Mathematics) from The Maharaja Sayajirao University of Baroda with Second Class (56.75%).

**Awards and Recognitions:**

Won international Earg Award in the category of excellence in research and development, presented by MathTech Thinking Foundation, Fazilka, India (A non-profit, Government of India approved NGO) On 19<sup>th</sup> March, 2024.

**According to Google Scholar**

**H-Index: 09**

**i10-Index: 09**

**Total Citations: 372**

**Publications:**

1. Ankita Jangid, B. S. Ratanpal and K. K. Venkataratnam, “Models of Anisotropic Stars in Spherically Symmetric Space-Time, *Grav. and Cosmo.*, **29**, (2023), 479.

2. Jaita Sharma, B. S. Ratanpal, Shivam Munshi and Vishant Shah, "Trajectory Controllability of Artificial Satellite Under the Effect of Oblateness of Earth", *Tuijin Jishu/J. Prop. Tech.*, **44** (2023) 3782.
3. Jaita Sharma, B. S. Ratanpal, Shivam Munshi and Vishant Shah, "Controllability Analysis of Motion of Artificial Satellite Under the Effect of Oblateness of Earth", *Tuijin Jishu/J. Prop. Tech.*, **44** (2023) 5381.
4. Rinkal Patel, B. S. Ratanpal and D. M. Pandya, "New Charged Anisotropic Solution on Paraboloidal Spacetime", *Astrophys. Space Sci.*, **368**, (2023), 58. **(Citations: 02)**
5. Ankita Jangid, B. S. Ratanpal and K. K. Venkataratnam, "Anisotropic Compact Star Model on Finch-Skea Space-Time", *Grav. and Cosmo.*, **29**, (2023), 206.
6. B. S. Ratanpal and Rinkal Patel, "Anisotropic Approach: Compact Star as a Generalized Model", *Astrophys. Space Sci.*, **368**, (2023), 21.
7. B. S. Ratanpal, V. O. Thomas and Rinkal Patel, "Compact Relativistic Star Under Karmakar Condition", *New Astronomy*, **100**, (2023), 101970. **(Citations: 01)**
8. B. S. Ratanpal, "Cracking and Stability of Non-rotating Relativistic Spheres with Anisotropic Internal Stresses", *IOP SciNotes*, **1**, (2020), 025207. **(Citations: 05)**
9. Jaita Sharma, B. S. Ratanpal, U. M. Pirzada and Vishant Shah, "Motion of Satellite under the Effect of Oblateness of Earth and Atmospheric Drag", *Int. J. Analy. Exper. Model Analy.*, **XI**, (2019) 2514. **(Citations: 01)**
10. B. S. Ratanpal and B. Bhar, "A New Class of Anisotropic Charged Compact Stars", *Phys. Astro. Int. J.*, **1**, (2017), 1. **(Citations: 10)**
11. S. K. Maurya, B. S. Ratanpal and M. Govender, "Anisotropic stars for spherically symmetric spacetimes satisfying the Karmarkar condition", *Annals of Physics*, **382**, (2017), 36. **(Citations: 68)**
12. B. S. Ratanpal, D. M. Pandya, R. Sharma and S Das, "Charged compact stellar model in Finch-Skea spacetime", *Astrophys. Space Sci.*, **362**, (2017) 82. **(Citations: 10)**
13. P. Bhar and B. S. Ratanpal, "A new anisotropic compact star model having Matese & Whitman mass function", *Astrophys. Space Sci.*, **361**, (2016) 217. **(Citations: 15)**
14. Monali Dodia, Vaibhavi Khatri, Nikul Patel and B. S. Ratanpal, "Determination of Velocity Air Distribution in a room for Central Air Conditioning System using Mathematical Model", *International Journal of Fluid and Thermal Engineering*, **5**, (2016) 16.
15. B. S. Ratanpal and Jaita Sharma, "Charged anisotropic star on paraboloidal space-time", *Pramana Journal of Physics*, **86**, (2016) 527. **(Citations: 05)**
16. B. S. Ratanpal, V. O. Thomas and D. M. Pandya, "Anisotropic star on pseudo-spheroidal spacetime", *Astrophys. Space Sci.*, **361** (2016) 65. **(Citations: 19)**
17. B. S. Ratanpal, V. O. Thomas and D. M. Pandya, "A new class of solutions of anisotropic charged distributions on pseudo-spheroidal spacetime", *Astrophys. Space Sci.*, **360** (2015) 53. **(Citations: 13)**

18. B. S. Ratanpal, Som Sahni, “On Speech Synthesis of Sindhi Numeric”, *Indian Journal of Science and Technology*, **8** (2015). **(Citations: 04)**

19. Jaita Sharma, B. S. Ratanpal, U. M. Pirzada, V. S. Shah, A. B. Chavda, Nirav Dave, “Study of Effect of Perturbation due to Oblateness of the Earth on Satellite”, Proceedings of 19th Annual Cum 4<sup>th</sup> International Conference of Gwalior Academy of Matematical Sciences (GAMS), Sardar Vallabhbhai National Institute of Technology, Surat, October 3 6, (2014).

20. B. S. Ratanpal, S. Ramamohan and V. O. Thomas, “ON MODELS OF STARS ON PARABOLOIDAL SPACETIME”, *Bulletin of the Marathwada Mathematical Society*, **14**, (2013) 51.

21. R. Sharma and B. S. Ratanpal, “RELATIVISTIC STELLAR MODEL ADMITTING A QUADRATIC EQUATION OF STATE”, *Int. J. Mod. Phys. D.*, **22** (2013) 1350074. **(Citations: 116)**

22. V. O. Thomas and B. S. Ratanpal, “NON-ADIABATIC GRAVITATIONAL COLLAPSE WITH ANISOTROPIC CORE”, *Int. J. Mod. Phys. D.*, **16** (2007) 1479. **(Citations: 22)**

23. V. O. Thomas, B. S. Ratanpal and P. C. Vinodkumar, “CORE-ENVELOPE MODELS OF SUPERDENSE STAR WITH ANISOTROPIC ENVELOPE”, *Int. J. Mod. Phys. D.*, **14** (2005) 85. **(Citations: 64)**

#### **Work Experience:**

1. Assistant Professor, Department of Applied Mathematics, Faculty of Technology & Engineering, The Maharaja Sayajirao University of Baroda, Vadodara from 24/07/2009 to present.

2. Lecturer, Babaria Institute of Technology, Varnama, Vadodara from 1/8/2006 to 19/7/2009.

3. Temporary Teaching Assitant, Department of Applied Mathematics, Faculty of Technology & Engineering, The Maharaja Sayajirao University of Baroda, Vadodara from Dec 2001 to July 2006.

#### **Invited Lectures and Chairmanships at Regional / State / National or International Conference / Seminars:**

| Sr. No. | Title of Lecture/ Academic Session | Title of Conference/Seminar etc                           | Date(s) of the event     | Organized by                                                                | Whether International/ National/State |
|---------|------------------------------------|-----------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------|---------------------------------------|
| 1.      | Data Communication                 | Seminar on Computer Networking, Apache Server and MySql   | 06-01-2009 to 09-01-2009 | Department of Mathematics, Indian Institute of Space Science and Technology | Institute                             |
| 2.      | Basics of MATLAB                   | Recent Trends in Computational Techniques for Engineering | 01-05-2012 to 05-05-2012 | Lukhdhirji Engineering College, Morbi                                       | Regional                              |
| 3.      | MATLAB Training                    | Training Program on Pedagogy of Engineering Mathematics   | 24-06-2013 to 28-06-2013 | Parul Institute of Technology, Limbda                                       | Regional                              |
|         |                                    |                                                           |                          |                                                                             |                                       |

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|-----|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------|---------------|
| 4.  | Basics of MATLAB and MATLAB Programming             | Mathematical, Statistical, Operation Research based Modeling and Simulation for Researchers, Engineers and Scientists | 27-01-2014 to 31-01-2014        | Department of Applied Mathematics & Humanities, SVNIT, Surat                                                     | National      |
| 5.  | Programming in MATLAB                               | Advanced Analytical & Numerical Techniques for Engineers and Scientist                                                | 03-03-2014 to 7-03-2014         | Department of Applied Mathematics & Humanities, SVNIT, Surat                                                     | National      |
| 6.  | Basics of LaTeX                                     | A State Level Workshop on “LaTeX – An Efficient Documentation Tool                                                    | 28 <sup>th</sup> December, 2014 | Department of Electrical Engineering, Polytechnic, The Maharaja Sayajirao University of Baroda                   | Regional      |
| 7.  | Programming Techniques in MATLAB                    | One Week Instructional Workshop on Scientific Computing using MATLAB                                                  | 07-01-2015 to 11-01-2015        | Department of Applied Mathematics, Faculty of Technology & Engineering, The M. S. University of Baroda, Vadodara | Regional      |
| 8.  | Problem Solving Techniques in Calculus of Variation | National Workshop on Problem Solving Techniques in Mathematics                                                        | 21-02-2015 to 26-02-2015        | Department of Mathematics, Faculty of Science, The Maharaja Sayajirao University of Baroda                       | National      |
| 9.  | Coordinator                                         | Instructional Workshop on Scientific Computing using MATLAB                                                           | 07-01-2015 to 11-01-2015        | Department of Applied Mathematics, Faculty of Technology & Engineering, The M. S. University of Baroda, Vadodara | Regional      |
| 10. | Session Chair                                       | International Conference on Natural Language Processing in Regional Languages                                         | 18-09-2015 to 19-9-2015         | Department of Computer Science, Punjabi University, Patiala                                                      | International |
| 11. | Simulation Techniques Applied in Real Life Problems | Short Term Training Program on Mathematical Modelling and Simulation for Researchers, Engineers and Scientists        | 11-07-2016 to 15-07-2016        | Department of Applied Mathematics and Humanities, SVNIT, Surat                                                   | National      |
|     |                                                     |                                                                                                                       |                                 |                                                                                                                  |               |

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| 12. | On Equation of Continuity                                                        | Short Term Training Program on Recent Advances in Computational Fluid Dynamics                                    | 13-03-2017 to 17-03-2017 | Department of Applied Mathematics and Humanities, SVNIT, Surat                                                                                                      | National |
| 13. | Problem Solving Techniques in Calculus of Variation                              | National Workshop on Problem Solving Techniques in Mathematics – II                                               | 27-11-2017 to 01-12-2017 | Department of Mathematics, Faculty of Science, The Maharaja Sayajirao University of Baroda                                                                          | National |
| 14. | An Introduction to Calculus of Variation                                         | Short Term Training Program on Recent Trends in Optimization Techniques & Applications in Science and Engineering | 04-12-2017 to 09-12-2017 | Vadodara Institute of Engineering, Kotambi, Vadodara – 391 510                                                                                                      | National |
| 15. | MATLAB for Mathematicians and Engineers                                          | Training Workshop on Computing Mathematics using MATLAB                                                           | 24-04-2021 to 30-04-2021 | MathTech Thinking Foundation, A ministry of corporate affairs, Government of India Approved NGO                                                                     | National |
| 16. | LaTeX an Excellent Documentation Tool                                            | Training Virtual Workshop on Latex and Statistical Analysis Using Matlab                                          | 22-05-2021 to 28-05-2021 | MathTech Thinking Foundation, A ministry of corporate affairs, Government of India Approved NGO                                                                     | National |
| 17. | Motion of Satellite under the effect of oblateness of Earth and Atmospheric Drag | National Webinar on Computational Methods in Applied Mathematics and Engineering                                  | 13-08-2021 to 14-08-2021 | Department of Applied Science and Humanities, Sankalchand Patel College of Engineering (SPCE), affiliated to Sankalchand Patel University, Visnagar, Gujarat, India | National |
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| 18. | Applications of Non-linear Differential Equations in the Study of Suerdense Stars on Geometrically Significant Spacetimes | On-line Seminar                                                   | 16-09-2021               | Department of Mathematical Physics of the Institute of Mathematics of NAS of Ukrain, Kyiv, Ukrain | International |
| 19. | Numerical Solution of First Order IVP using MATLAB                                                                        | Numerical Methods in Engineering and Physical Sciences with MATAB | 04-06-2022 to 10-06-2022 | MathTech Thinking Foundation, A ministry of corporate affairs, Government of India Approved NGO   | National      |